

HW2.3

SCIE 001 Math

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1. Recall that the probability density function for the electron-proton distance r in the hydrogen atom ground state (i.e., the 1s orbital) is given by

$$\begin{aligned} f(r) &= 4\pi r^2 \psi_{100}^2(r) \\ &= A r^2 e^{-\frac{2r}{a_0}} \end{aligned}$$

where a_0 is the Bohr radius.

- (a) Find the cumulative distribution function $F(r)$.
- (b) Determine the value of A (i.e., “normalize” the distribution).
- (c) Determine the probability that the electron will be found within $2a_0$ of the proton. Does this seem reasonable?
- (d) Determine the “most probable” and the “expected” radius for the hydrogen atom in the ground state.

2. *Newton’s universal law of gravitation* states that the force of attraction between two point masses m and M has magnitude

$$F = \frac{GmM}{r^2}$$

where r is the distance between the masses and $G = 6.67 \times 10^{-11} \text{ N} \cdot \frac{\text{m}^2}{\text{kg}^2}$ is a constant.

- (a) If M represents the mass of the centre of the earth and we regard it as a point mass concentrated at its centre, show that Newton’s universal law of gravitation at the earth’s surface reduces to $F = mg$, where $g = 9.82 \text{ m/s}^2$. Assume for the calculation of M that the earth is a sphere with radius 6370 km and mean density $5.52 \times 10^3 \frac{\text{kg}}{\text{m}^3}$.